

Adaptive and Coordinated Targeted Label Flipping Attacks in Federated Learning with Krum-Based Defense

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Abstract

Federated learning (FL) is a machine learning approach that allows multiple decentralized clients to collaboratively learn a shared model without sharing their training data. While this privacy-preserving approach enhances data privacy and reduces the risks associated with centralized data collection, it also introduces new security vulnerabilities in the model aggregation process. In particular, poisoning attacks performed by malicious clients can affect a global model by corrupting training behavior. This remains difficult to detect under realistic non-IID data distributions. In this research, we study label flipping attacks against federated learning using the Federated Averaging (FedAvg) aggregation algorithm. Initially, we perform a basic attack where malicious clients relabel samples belonging to a source class into an attacker-chosen target class. We then progressively strengthen the attack by applying adaptive scaling strategies and coordinated multi-client poisoning to improve the Attack Success Rate (ASR) while preserving Main Task Accuracy (MTA). Experimental results show that adaptive and coordinated adversaries significantly increase poisoning effectiveness under a non-IID federated learning setting. To mitigate these attacks, we implement the Krum Byzantine-robust aggregation defense and evaluate its effectiveness against coordinated poisoning attacks. Our experiments on the MNIST dataset demonstrate that adaptive multi-client attacks can greatly boost ASR while maintaining acceptable clean-data accuracy under FedAvg aggregation. Nonetheless, the Krum defense effectively lowers ASR to levels approaching zero while maintaining high MTA, demonstrating the importance of robust aggregation mechanisms in federated learning systems.

1 Introduction

Federated Learning (FL) has evolved into the most significant decentralized machine learning paradigm for conducting collaborative learning in a privacy-preserving manner [1]. Federated learning allows training in a distributed manner across multiple clients while keeping data locally stored, unlike traditional centralized machine learning systems that collect raw data on a central server. Clients train localized models based on their individual datasets in isolation and share model updates with the server periodically. The updates are then aggregated on a central server.

The decentralized nature offers benefits including enhanced user privacy, reduced data transfer requirements, and improved data governance. Federated learning has garnered considerable interest in healthcare, finance, mobile applications, and cybersecurity [5].

Although there are benefits, federated learning has security issues. Because the server collects updates from different distributed

participants, a malicious client can harm the global model by poisoning local training [2]. In centralized systems, one is able to validate the training data; however, one cannot do so in federated learning systems. The server does not have direct access to client data and therefore must trust the client updates. This provides an opportunity for adversaries to introduce malicious behavior into the aggregation process.

An important sub-category of poisoning attacks is a targeted label flipping attack [4]. In this attack, malicious clients relabel samples from one class to another class during local training. The main idea is not to destroy the whole model but to induce systematic targeted misclassification behavior while maintaining good performance on clean inputs. These attacks can be difficult to detect while still introducing targeted malicious behavior.

Most existing studies focus on malicious behavior that is either static or isolated. Real-world federated learning deployments, however, may encounter more sophisticated adversaries that can adapt attacks over time or coordinate across multiple malicious clients [6]. In adaptive attacks, the poisoning strength may change throughout the training rounds. Besides, coordinated attacks by colluding clients may enhance malicious influences over the aggregation.

We study increasingly sophisticated targeted label flipping attacks in federated learning under non-IID client distributions. We initiate a baseline targeted poisoning attack using the FedAvg aggregation method. We then incrementally reinforce the attack by adapting scaling and introducing coordinated multi-client poisoning. In the end, we apply the Krum Byzantine-robust aggregation defense to test whether robust aggregation can suppress adaptive multi-client poisoning. The primary contributions of this project are:

- (1) Implementation of a baseline targeted label flipping attack under FedAvg aggregation.
- (2) Development of an adaptive stealth-oriented poisoning strategy using dynamic scaling.
- (3) Evaluation of coordinated multi-client poisoning attacks under non-IID FL settings.
- (4) Implementation of the Krum robust aggregation defense against poisoning attacks.
- (5) Comprehensive experimental analysis using Main Task Accuracy (MTA) and Attack Success Rate (ASR).
- (6) Investigation of the security-utility tradeoff between attack effectiveness and model usability.

2 Related Work

Federated learning is perhaps the most prominent decentralized machine learning paradigm that enables collaborative model training without directly sharing user data. McMahan et al. introduced

Federated Averaging [1], one of the first and most influential federated learning algorithms. FedAvg lets multiple clients train their local model discretely and send model updates to the central server at intervals, greatly enhancing privacy protection and reducing communication overhead.

In spite of the many advantages federated learning brings, poisoning attacks have evolved into a significant security concern, as the central server usually does not have access to the client datasets [3]. As a result, malicious clients can influence local training and introduce malicious updates into aggregation.

Label flipping attacks are among the easiest yet highly effective poisoning attacks in FL [4]. During local training, adversarial clients intentionally relabel training samples from one class to another. While reasonably easy to implement, these attacks can drastically change model behavior under distributed training environments.

Numerous research works have investigated adaptive poisoning and model poisoning attacks. According to Bhagoji et al. [2], model updates from malicious clients can be used to gain adversarial influence during aggregation via scaling strategies. Adaptive attacks are dangerous as they not only maximize the effectiveness of poisoning but also stay hidden from detection.

There is growing interest in coordinated poisoning attacks involving multiple malicious clients [6]. Rather than a single malicious participant, coordinated attackers work together to dominate the aggregation, significantly increasing ASR and making conventional aggregation strategies more vulnerable.

To prevent Byzantine and poisoning attacks, various aggregation algorithms like Krum were proposed [3]. Krum protects federated learning systems by choosing model updates that stay near the majority of participating clients, identifying suspicious outlier updates and reducing their influence on global model aggregation.

Earlier studies mainly studied unrelated or static adversarial mechanisms. Meanwhile, our work studies progressively adaptive and coordinated multi-client poisoning attacks in non-IID federated learning settings and assesses the Krum Byzantine-robust aggregation defense's effectiveness.

3 Methodology

3.1 Federated Learning Architecture

The project implements a federated learning environment consisting of several distributed clients and a central aggregation server. Each client independently trains a private partition of a local convolutional neural network (CNN) without sharing raw training samples [1].

At the start of each communication round, the server shares the up-to-date global model with all clients. Clients then perform local training on their own datasets and send model parameters back to the server. The updated global model is created by aggregating local updates on the server through the FedAvg algorithm. The overall workflow is:

- (1) Set up the global CNN model.
- (2) Distribute the MNIST dataset among clients using non-IID distribution.
- (3) Share the global model among all clients.
- (4) Have each client perform local training.
- (5) Gather local model updates at the server.

- (6) Utilize FedAvg or Krum to combine updates.
- (7) Update the global model.
- (8) Use MTA and ASR metrics to evaluate model performance.
- (9) Repeat for multiple rounds of communication.

3.2 Dataset and Non-IID Distribution

All experiments are conducted using the MNIST handwritten digit dataset [7], consisting of grayscale images of handwritten digits from zero to nine. A realistic federated learning environment is simulated by distributing the dataset across clients using a non-IID partitioning scheme. The non-IID setting is implemented as follows:

- Different clients have different digit distributions.
- Some clients might have limited exposure to specific classes.
- Training becomes more diverse and challenging.

The non-IID setting enhances realism since malicious updates become harder to distinguish from naturally diverse benign updates.

3.3 CNN Model Architecture

All clients utilize a compact convolutional neural network (CNN) framework for local training, designed for MNIST image classification. The model consists of convolutional layers for feature extraction, ReLU activation functions, pooling layers for dimensionality reduction, fully connected layers for classification, and a Softmax output layer for digit prediction. The CNN model is synchronized at all clients in each communication round.

3.4 Threat Model

3.4.1 Purpose of the Adversary. The adversary aims to carry out targeted misclassification attacks without much degradation in overall model accuracy. Malicious clients try to make the global model incorrectly classify digit 1 images as digit 7. To stay undetected, the adversary maximizes ASR while maintaining minimum MTA.

3.4.2 Adversary Capability. The adversary controls one or more malicious FL clients capable of changing local training labels, compromising local datasets, altering local model updates, coordinating attacks across multiple malicious participants, and implementing adaptive scaling strategies. Malicious participants behave like benign clients, making detection difficult.

3.4.3 Adversary Knowledge. The adversary is assumed to know the global model architecture, the aggregation algorithm, communication round information, and the federated training configuration. However, malicious clients do not have direct access to other benign client datasets.

3.4.4 Realism of the Threat Model. This threat model is realistic because federated learning systems inherently trust client updates [2]. Federated learning devices can be compromised due to malware, insider attacks, or adversarial software modifications. Real-world scenarios include compromised mobile devices, IoT systems, healthcare organizations with competing stakeholder interests, and collaborative training environments with insider attackers.

3.5 Baseline Targeted Label Flipping Attack

This project first implements a baseline targeted label flipping attack [4]. Malicious clients relabel training samples from source class (digit 1) to target class (digit 7) during local training. The objective is not to break the entire model, but to create targeted misclassification while maintaining good performance on clean testing samples. Only malicious client datasets receive the poisoning attack, while benign clients train normally.

3.6 Adaptive Poisoning Strategy

Adaptive scaling techniques are proposed to improve the poisoning attack [2]. Malicious clients gradually increase their attack intensity during communication rounds using two adaptable parameters.

3.6.1 Stealth Weight. The stealth weight determines the magnitude of deviation between malicious updates and benign updates during early training rounds. Less noticeable alterations make attacks on initial aggregations less detectable.

3.6.2 Boost Factor. The boost factor increasingly enhances malicious model updates in subsequent communication rounds. As training continues, malicious updates exert increasing strength over global aggregation.

The adaptive strategy allows attackers to maintain low-magnitude updates early, gradually increase poisoning effectiveness, and improve ASR while preserving acceptable MTA. This simulates a realistic adversary capable of adjusting attack intensity dynamically throughout federated training.

Algorithm 1: Adaptive PGD Label Flipping Attack

Input: Global model $W^{(r)}$, source class $s=1$, target class $t=7$,
 $\lambda_{\text{start}}=2.0$, $\lambda_{\text{end}}=0.2$, $\beta_{\text{start}}=2.0$, $\beta_{\text{end}}=4.5$, $R=20$
rounds, $E=3$ epochs, $\text{lr} \eta=0.01$

Output: Updated global model $W^{(R)}$

```

1 for  $r \leftarrow 0$  to  $R - 1$  do
2    $\lambda_r \leftarrow \lambda_{\text{start}} + (\lambda_{\text{end}} - \lambda_{\text{start}}) \cdot \frac{r}{R-1}$ ;
3    $\beta_r \leftarrow \beta_{\text{start}} + (\beta_{\text{end}} - \beta_{\text{start}}) \cdot \frac{r}{R-1}$ ;
4   for each client  $i \in \{0, \dots, N-1\}$  do
5     if client  $i$  is malicious then
6       Flip labels:  $\tilde{y} \leftarrow t$  for all  $(x, y) \in D_i$  where  $y=s$ ;
7        $W_i \leftarrow W^{(r)}$ ;
8       for epoch  $e \leftarrow 1$  to  $E$  do
9          $\mathcal{L} \leftarrow \text{CE}(W_i; \tilde{D}_i) + \lambda_r \sum_{\ell} (W_i^{\ell} - W^{(r)\ell})^2$ ;
10         $W_i \leftarrow W_i - \eta \nabla_{W_i} \mathcal{L}$ ;
11         $\hat{W}_i \leftarrow W^{(r)} + \beta_r (W_i - W^{(r)})$ ; // boosted
           update
12      else
13        Train  $W_i$  on clean  $D_i$  for  $E$  epochs via SGD;
14         $\hat{W}_i \leftarrow W_i$ ;
15   $W^{(r+1)} \leftarrow \frac{1}{N} \sum_{i=0}^{N-1} \hat{W}_i$ ; // FedAvg
```

3.7 Coordinated Multi-Client Poisoning

This project also studies coordinated poisoning where multiple colluding malicious clients launch an attack simultaneously [6].

Three configurations are analyzed: 1, 3, and 5 malicious clients out of 10 total. During local training, each malicious client performs synchronized poisoning. When attackers coordinate, they together enhance their harmful influence during aggregation, resulting in a more effective attack than a single malicious client. With an increase in malicious clients, the likelihood of attack success is higher and malicious updates dominate honest updates more efficiently.

3.8 FedAvg Aggregation

The Federated Averaging (FedAvg) algorithm [1] is used as the baseline aggregation mechanism, averaging all participating model parameters equally. The global update process is:

$$W_{\text{global}} = \frac{1}{N} \sum_{i=1}^N W_i \quad (1)$$

where W_i represents the local model update from client i , and N represents the number of participating clients. Although computationally efficient, FedAvg remains vulnerable to poisoning attacks because all client updates contribute equally during aggregation.

3.9 Krum Byzantine-Robust Aggregation Defense

The Krum Byzantine-robust aggregation algorithm [3] is deployed to resist coordinated poisoning attacks. Unlike FedAvg, Krum calculates pairwise Euclidean distances between model updates and selects the update closest to the majority of participating clients. This helps reduce the influence of malicious or abnormal updates during aggregation. Because poisoned updates often behave as outliers, Krum mitigates malicious influence by selecting updates that better represent the majority of honest clients.

Algorithm 2: Krum Byzantine-Robust Aggregation

Input: Local model updates $\{W_i\}_{i=0}^{N-1}$, number of malicious clients f

Output: Selected global model update W^*

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1 Flatten each  $W_i$  into vector  $v_i = \text{vec}(W_i)$ ;
2 for each client  $i \leftarrow 0$  to  $N - 1$  do
3   for each client  $j \neq i$  do
4      $d_{ij} \leftarrow \|v_i - v_j\|^2$ ;
5   Sort  $\{d_{ij}\}_{j \neq i}$  in ascending order;
6    $k \leftarrow N - f - 2$ ;
7    $\text{score}(i) \leftarrow \sum_{\text{top-}k} d_{ij}$ ; // sum of  $k$  smallest
           distances
8  $i^* \leftarrow \arg \min_i \text{score}(i)$ ;
9  $W^* \leftarrow W_{i^*}$ ;
10 return  $W^*$ ;
```

4 Experimental Setup

4.1 Environment Configuration

The experiments are implemented using Python, PyTorch, and Jupyter Notebook, with GPU acceleration when available.

4.2 Experimental Parameters

Table 1 summarizes the experimental parameters used across all experiments.

Table 1: Experimental Parameters

Parameter	Value
Dataset	MNIST (60k train / 10k test)
Number of Clients	10
Communication Rounds	20
Local Epochs	3
Batch Size	32
Learning Rate	0.01
Aggregation	FedAvg / Krum
Poison Ratio	Varied per attack configuration
Distribution Type	Non-IID

4.3 Evaluation Metrics

4.3.1 Main Task Accuracy (MTA). MTA measures the classification accuracy of the global model on clean testing samples. A high MTA indicates the model is still performing the intended classification task effectively despite the presence of malicious clients. Maintaining a high MTA was important because successful poisoning attacks should ideally remain stealthy without significantly degrading overall model performance.

4.3.2 Attack Success Rate (ASR). ASR measures the percentage of source-class samples that are incorrectly classified as the target class after the attack. A higher ASR indicates a stronger poisoning attack. If MTA drops significantly, the attack becomes easier to detect.

4.3.3 Security–Utility Tradeoff. The relationship between MTA and ASR demonstrates the security–utility tradeoff in FL. An attack achieving high ASR while maintaining high MTA is considered more dangerous. The effectiveness of the Krum defense was evaluated based on its ability to reduce ASR while preserving acceptable MTA.

5 Experimental Results and Analysis

This section presents the findings of the experiments conducted across different poisoning attack scenarios in a non-IID federated learning environment using the MNIST dataset. The efficacy of attacks and defense methods was assessed through MTA and ASR.

5.1 Baseline Label Flipping Attack

The baseline experiment used 2 malicious clients out of 10 total, with a poison ratio of 1.0 and an attack scale of 5.0, flipping all source-class (digit 1) labels to the target class (digit 7) during local training. Under FedAvg aggregation, the attack reached a final ASR of 0.9683 and a final MTA of 0.8715 after 20 communication rounds. The high ASR confirms that FedAvg’s equal-weight averaging provides no protection against scaled malicious updates: with only 2 of 10 clients poisoning, the global model learned to misclassify nearly all digit-1 test samples as digit 7. The MTA drop to 0.8715 reflects collateral damage to overall accuracy, indicating that a scale factor

of 5.0 is aggressive enough to partially distort non-targeted classes as well. This experiment establishes the reference point against which the more sophisticated adaptive and stealthy attacks are compared.

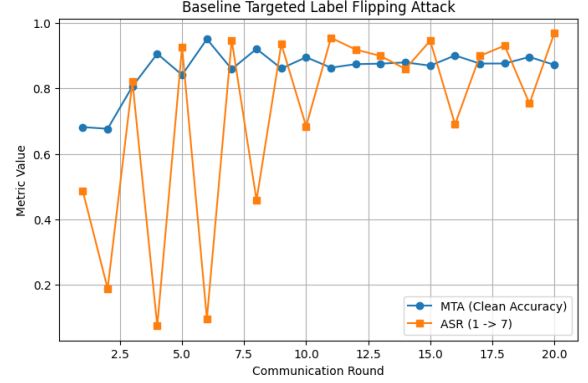


Figure 1: Baseline attack: MTA and ASR over 20 rounds (2 malicious clients, scale 5.0).

5.2 Adaptive Poisoning Attack

The adaptive attack used 3 malicious clients with PGD-constrained local training, where a stealth weight penalized large deviations from the global model and a boost factor amplified the final update delta. Both parameters were scheduled across 20 communication rounds: the stealth weight decayed from 2.0 to 0.2 while the boost factor increased from 2.0 to 4.5, transitioning the attack from a low-visibility phase in early rounds to a high-impact phase in later rounds. This produced a final ASR of 0.9868 and a final MTA of 0.8549 under FedAvg aggregation. The stealthy two-phase variant (4 malicious clients, clean pre-training followed by poison fine-tuning, scale 1.6) achieved a final ASR of 0.7374 with MTA of 0.9004, demonstrating that restricting the magnitude of malicious updates preserves clean accuracy at the cost of a lower ASR ceiling. Together, these results show that adaptive scheduling meaningfully increased poisoning effectiveness over the static baseline while maintaining acceptable model utility for the attacker.

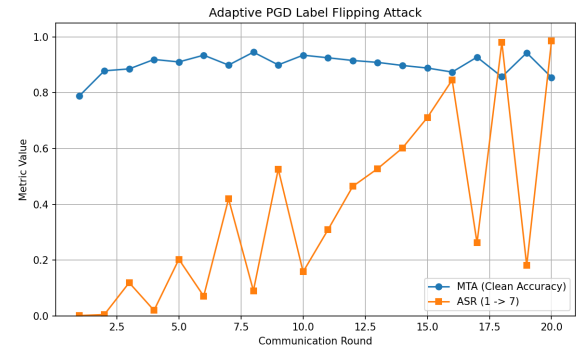


Figure 2: Adaptive PGD attack: MTA and ASR over 20 rounds under FedAvg (3 malicious clients, stealth weight 2.0→0.2, boost factor 2.0→4.5).

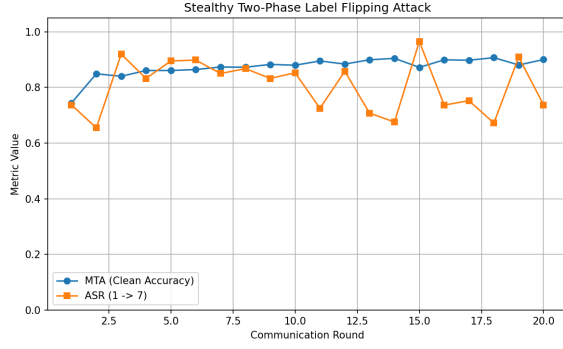


Figure 3: Stealthy two-phase attack: MTA and ASR over 20 rounds (4 malicious clients).

5.3 Coordinated Multi-Client Poisoning Attack

Three configurations were tested under FedAvg aggregation using the adaptive PGD attack: 1, 3, and 5 malicious clients out of 10. Table 2 summarizes the final-round results.

Table 2: Multi-Client Adaptive Attack Results under FedAvg (20 rounds, MNIST, class 1 $\rightarrow$ 7)

Malicious Clients	Final MTA	Final ASR
1	0.9792	0.0062
3	0.8894	0.7665
5	0.8773	0.5753

With 1 malicious client, the attack was largely ineffective, reaching only ASR = 0.0062 while MTA remained high at 0.9792. Increasing to 3 malicious clients produced a dramatic shift: ASR rose to 0.7665 with MTA at 0.8894, crossing the tipping point at which coordinated poisoning begins to dominate aggregation. With 5 malicious clients, ASR was 0.5753 and MTA was 0.8773. Across all configurations, MTA remained above 0.87, indicating the attacks stayed sufficiently stealthy to avoid detection through accuracy monitoring alone. These results confirm that coordinated poisoning under FedAvg is a serious threat once the number of colluding clients exceeds roughly 20–30% of the total.

5.4 Krum Defense Evaluation

Having established ASR baselines under FedAvg, the same adaptive multi-client attack was re-run with Krum replacing FedAvg as the aggregation algorithm [3]. The Krum defense selects a single client update per round — the one with the smallest sum of distances to its nearest neighbors — which makes it structurally resistant to outlier updates generated by malicious clients. Table 3 shows the results.

The results under Krum were decisive across all three configurations. ASR dropped to 0.0009 in every case — even with 5 malicious clients (50% of the federation) — while MTA remained above 0.94. In all cases the attack was completely neutralized. Notably, MTA under Krum was higher than under FedAvg in every configuration, since Krum’s selection of the best-fit update tends to pick the cleanest update rather than averaging in malicious noise. The contrast

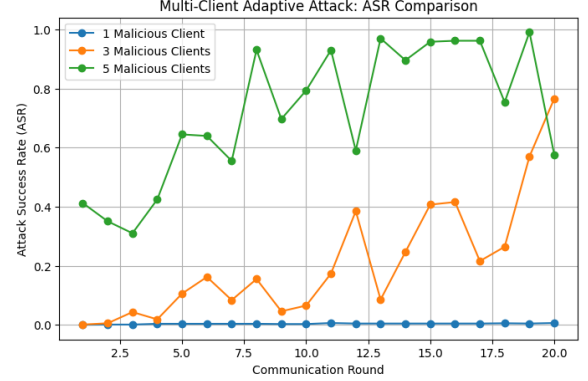


Figure 4: Multi-client attack ASR over 20 rounds for 1, 3, and 5 malicious clients.

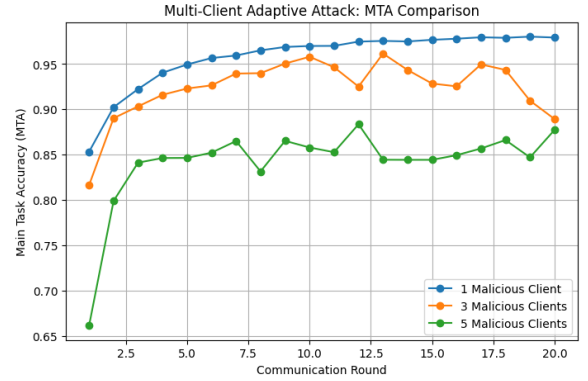


Figure 5: Multi-client attack MTA over 20 rounds for 1, 3, and 5 malicious clients.

Table 3: Krum Defense Results under the Adaptive Multi-Client Attack (20 rounds, MNIST, class 1 $\rightarrow$ 7)

Malicious Clients	Final MTA	Final ASR
1	0.9749	0.0009
3	0.9723	0.0009
5	0.9448	0.0009

between FedAvg (ASR up to 0.9683 with 2 malicious clients) and Krum (ASR near 0.0009 even with 5 malicious clients) demonstrates the importance of Byzantine-robust aggregation as a first line of defense against label flipping attacks.

5.5 Discussion

The results indicate that poisoning attacks are still a significant security concern in federated learning. Even basic label flipping attacks impacted the global model when using FedAvg aggregation. More sophisticated adaptive attacks further enhanced poisoning effectiveness by steadily amplifying adversarial behavior across communication rounds.

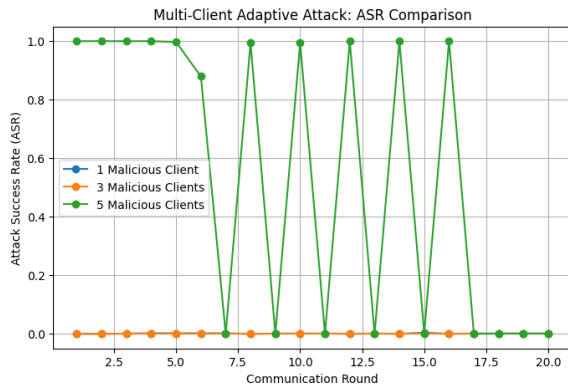


Figure 6: Krum defense: ASR over 20 rounds for 1, 3, and 5 malicious clients. With 1 and 3 malicious clients ASR is suppressed to near zero; with 5 malicious clients (50% of the federation) the defense shows instability throughout training but recovers by round 20.

The coordinated multi-client poisoning experiments demonstrated that colluding malicious clients can greatly amplify the attack. Traditional averaging-based aggregation methods are susceptible to collaborative adversarial settings. The experiments also revealed the relationship between MTA and ASR: attacks that maintained a high MTA remained hidden because the model continued to function well on regular data while introducing targeted malicious behavior.

The Krum defense demonstrates that robust aggregation can diminish the impact of malicious updates. Nonetheless, Byzantine-robust aggregation schemes may fail when the count of malicious clients becomes too high or attack strategies become even more adaptive. Overall, secure aggregation mechanisms are essential in federated learning systems, especially when clients are distributed non-IID.

6 Conclusion and Future Work

Through this work, we studied targeted label flipping poisoning attacks in federated learning under non-IID client distributions. We first introduced a baseline poisoning attack using the FedAvg aggregation algorithm and then gradually strengthened the attack using dynamic scaling strategies and coordinated multi-client attacks.

The experimental results demonstrate that coordinated, adaptive attacks increased ASR while not sacrificing MTA, and that colluding adversary clients can create severe vulnerabilities in standard federated aggregation functions. To protect against these attacks, the Krum Byzantine-robust aggregation algorithm was implemented and evaluated. Krum successfully reduced ASR to near-zero while ensuring high MTA across all configurations, highlighting how essential robust aggregation techniques are for protecting federated learning systems against poisoning attacks.

Federated learning systems remain susceptible to increasingly complex adversarial behavior, especially when data is non-IID. Simultaneously, robust aggregation defenses can significantly enhance the resilience of models against malicious participants.

Future research may consider involving other datasets such as CIFAR-10 or Fashion-MNIST to evaluate the poisoning behavior in more complex learning scenarios. Research can also be extended to backdoor attacks and gradient manipulation attacks. Moreover, Krum may be applied and compared against other Byzantine-robust defenses such as Trimmed Mean [8], Bulyan, or FLTrust to further enhance the security of federated learning.

7 Use of AI

AI tools were used in a limited and supplementary capacity during this project. Specifically, AI assistance was used to polish the written report by correcting grammar and improving sentence clarity, to assist with LaTeX formatting and resolving compilation errors in Overleaf, and to help debug experiments when optimizing for better Attack Success Rate (ASR) values during implementation. All experimental design, code architecture, data analysis, and technical conclusions are entirely the original work of the author. AI was not used to generate results, derive findings, or produce any core technical contributions.

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